

SERGIO KEVIN PEREZ NATEROS

Backend Developer | Python · FastAPI · Django · Process Automation

Legacy System Integration · Healthcare Domain

Tingo Maria, Peru • Remote worldwide • UTC-5 (full-day overlap with US time zones)

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PROFESSIONAL SUMMARY

Backend developer with 5 years building and shipping production automation systems inside a high-volume public hospital, backed by 13 years of hands-on experience in healthcare operations. That combination is the differentiator: deep knowledge of how clinical and administrative workflows actually run, plus the engineering skills to turn them into software. Specialized in integrating closed legacy systems with no public API by reverse-engineering their HTTP traffic, and in replacing manual bureaucratic processes with automated pipelines that cut turnaround time and human error. Core stack: Python (FastAPI, Django), PostgreSQL, HTMX and Playwright. Full ownership end to end — requirements, architecture, implementation, deployment and end-user support.

PROFESSIONAL EXPERIENCE

EsSalud – Hospital I Tingo Maria (Peru's National Social Health Insurance)

Apr 2013 – Present

Tingo Maria, Peru — public hospital network serving the Huanuco region

Backend Developer / Systems Analyst – Patient Referrals Unit

2021 – Present

- Automated the end-to-end medical referral workflow: the system pulls referrals issued by the attending physician from the institutional platforms, consolidates them for the referral officer's approval, and flags inconsistencies in clinical notes and ICD-10 diagnostic coding before submission — cutting rejected records and rework between facilities.
- Own the rejected-referral remediation cycle across multiple medical specialties, including routing decisions between hospitals of different care levels, keeping patient continuity of care intact.
- Designed and shipped internal Python automation systems now running in production, replacing manual processes previously handled by clinical and administrative staff.
- Integrated with three closed institutional platforms that expose no public API, by analyzing HAR captures and reverse-engineering their HTTP traffic; implemented persistent sessions with keep-alive pings to prevent server-side session lockouts.
- Processed 15 months of historical referral and physician-scheduling data programmatically; the resulting technical reports supported the CT scanner procurement case before management and exposed a 399% demand-to-capacity gap in General Medicine.
- Ran the statistical analysis proving the facility's elevated appointment-deferral indicator was driven by follow-ups and inter-consultations rather than new demand, correcting how the metric was interpreted institutionally and the corrective actions taken.
- Automated executive reporting (Excel, Word, PowerPoint) with Python, replacing manually assembled monthly and quarterly technical reports.

Healthcare Data & Operations Assistant – Multi-department rotation

2013 – 2021

- Recorded and validated clinical and administrative data across the hospital's core information systems, safeguarding the integrity of the data feeding facility-level performance indicators.
- Rotated across multiple hospital departments as operational needs required, gaining end-to-end visibility into admissions, physician scheduling, patient referrals and statistical reporting.
- That cross-departmental process knowledge is what later made it possible to identify the real bottlenecks and design the automation systems the facility runs on today.

Independent Software Developer & Consultant

2024 – Present

Custom software for healthcare and retail clients

Remote

- Designed a pharmacy point-of-sale and inventory platform featuring national drug-registry lookup, cash management, customer loyalty, and FEFO lot tracking with expiry control.
- Built a middleware layer allowing an HL7-based clinic to query and operate an AMIS-850 medication dispenser that ships with a proprietary SDK only and no native HL7 support.
- Built FotoStudio Pro, an e-commerce platform for a professional photography studio: watermarked catalog, shopping cart, mobile-wallet (Yape) payment and digital delivery of high-resolution photos.
- Handle the full consulting cycle: discovery, technical proposal, pricing, delivery and support.

SELECTED PROJECTS

CitaExpress — Medical referral delivery middleware

FastAPI, SQLite, Jinja2, HTMX, TailwindCSS, ReportLab, httpx

Scrapes scheduled referrals from a closed hospital platform, generates stamped PDFs and delivers them to patients over a WhatsApp link, removing unnecessary trips to the hospital. Built on a persistent httpx session with keep-alive pings to avoid lockouts on the source system.

PediaClinic — Pediatric clinic management platform

Django 4.2, PostgreSQL, Celery, HTMX, Tailwind/Flowbite, ReportLab

Role-based platform for guardians, technicians, physicians and admins, with dual authentication (Google OAuth2 and internal credentials), mobile-wallet payments reconciled automatically over IMAP, reservation expiry via Celery Beat, Telegram bot notifications and PDF receipts.

FotoStudio Pro — Professional photography e-commerce

FastAPI, SQLite, Jinja2, TailwindCSS

Online store for a photography studio: watermarked photo catalog, shopping cart, mobile-wallet payment verification and digital delivery of high-resolution photographs.

Scheduling Robot — Browser automation with Playwright

Python, Playwright, openpyxl

Loads monthly staff rosters into an internal web form from Excel grids submitted by department heads, normalizing per-department routing rules and validating the 150-hour monthly target per employee.

Referral Bridge — Cross-institution referral generator

FastAPI, SQLAlchemy, SQLite, Jinja2, HTMX, ReportLab

Web app letting physicians issue institution-format referrals to an external imaging center — a workflow the hospital's own platform does not support natively.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML5, CSS3

Backend: FastAPI, Django, Django REST Framework, SQLAlchemy, Celery, httpx

Frontend: HTMX, Jinja2, TailwindCSS, Flowbite, Bootstrap

Databases: PostgreSQL, MySQL, SQL Server, SQLite

Automation: Playwright, web scraping, HAR analysis and API reverse-engineering, ReportLab, openpyxl

Data: Pandas, Power BI, statistical analysis, automated report generation

Integrations: HL7, OAuth2, IMAP/SMTP, REST APIs, Telegram bots, WhatsApp

Tools: Git, GitHub, Docker, GNU/Linux, Railway, Scrum

EDUCATION

Universidad Tecnológica del Peru (UTP)	2015 – 2021
Professional Engineering Degree — Systems & Computer Engineering	2026
<i>Titulo Profesional: post-bachelor thesis defended; licensed engineer in Peru</i>	
Bachelor's Degree, Systems & Computer Engineering	2021

CERTIFICATIONS & CONTINUING EDUCATION

12 accredited certifications — over 1,000 hours of technical training.

Platzi: Backend Development with Python track (Dec 2025); Professional Scrum (Dec 2025); Django REST Framework (Dec 2024); Frontend Developer (Jul 2022); HTML & CSS (Jul 2022)

Edutin Academy: Python Developer — 220 hrs (Dec 2025); Relational Databases: Design & Management — 180 hrs (Dec 2025); Cybersecurity & Information Security — 180 hrs (Dec 2025); GNU/Linux Systems Administration — 120 hrs (Dec 2025)

Data Science Analysis: Data Analysis with Python & Power BI — 40 hrs (2025); Python with SQL Server & MySQL for Big Data — 53 hrs (2025); Machine Learning with Python — 40 hrs (2025)

LANGUAGES

Spanish — Native • **English** — Intermediate (B1): technical reading and written communication; comfortable with async, written-first collaboration